

how the network/systems and applications (collectively referred to as the ecosystem) are performing with respect to pre-defined measures specific to time of day, day of week, or special events (e.g., storm surges or pay events, such as a boxing match). These pre-defined performance measures exist throughout the service path, from the customer's residence or business through the entire network, as well as the interface points to partners and peers. In addition, synthetic transactions are run to ensure the health of the ecosystem on a continual basis. **Fault Management** is defined as the ability to detect, log and understand anomalies that may impact customers. Comcast utilizes correlation engines to properly determine an event's severity and act appropriately, eliminating or remediating potential issues before they affect customers. **Configuration Management** makes sure appropriate versions of hardware and software are in place across all elements of the ecosystem. Keeping these elements at their peak "golden" levels helps them avoid unintended consequences. **Accounting Management** ensures that the operations centers have a clear understanding of the provisioning and utilization of the ecosystem. This is especially important to ensure that at all times the operations centers have the ability to re-route traffic effectively. **Security Management** ensures that the proper controls exist to ensure the ecosystem is effectively protected against inappropriate access.

Network Operations Centers and the ecosystem they support are not static. Engineering and Operations personnel are constantly re-evaluating the pre-defined performance measures and tools to ensure that the customers' expectations for operational excellence are met.

9.3 The Internet-Standard Management Framework

Contrary to what the name SNMP (Simple Network Management Protocol) might suggest, network management in the Internet is much more than just a protocol for moving management data between a management entity and its agents, and has grown to be much more complex than the word "simple" might suggest. The current Internet-Standard Management Framework traces its roots back to the Simple Gateway Monitoring Protocol, SGMP [RFC 1028]. SGMP was designed by a group of university network researchers, users, and managers, whose experience with SGMP allowed them to design, implement, and deploy SNMP in just a few months [Lynch 1993]—a far cry from today's rather drawn-out standardization process. Since then, SNMP has evolved from SNMPv1 through SNMPv2 to the most recent version, SNMPv3 [RFC 3410], released in April 1999 and updated in December 2002.

When describing any framework for network management, certain questions must inevitably be addressed:

- What (from a semantic viewpoint) is being monitored? And what form of control can be exercised by the network administrator?
- What is the specific form of the information that will be reported and/or exchanged?
- What is the communication protocol for exchanging this information?

Recall our human organizational analogy from the previous section. The boss and the branch managers will need to agree on the measures of activity, productivity, and budget used to report the branch office's status. Similarly, they'll need to agree on the actions the boss can take (for example, cut the budget, order the branch manager to change some aspect of the office's operation, or fire the staff and shut down the branch office). At a lower level of detail, they'll need to agree on the form in which this data is reported. For example, in what currency (dollars, euros?) will the budget be reported? In what units will productivity be measured? While these may seem like trivial details, they must be agreed upon, nonetheless. Finally, the manner in which information is conveyed between the main office and the branch offices (that is, their communication protocol) must be specified.

The Internet-Standard Management Framework addresses the questions posed above. The framework consists of four parts:

- Definitions of *network management objects*, known as MIB objects. In the Internet-Standard Management Framework, management information is represented as a collection of managed objects that together form a virtual information store, known as the Management Information Base (MIB). An MIB object might be a counter, such as the number of IP datagrams discarded at a router due to errors in an IP datagram header, or the number of carrier sense errors in an Ethernet interface card; descriptive information such as the version of the software running on a DNS server; status information such as whether a particular device is functioning correctly; or protocol-specific information such as a routing path to a destination. MIB objects thus define the management information maintained by a managed device. Related MIB objects are gathered into **MIB modules**. In our human organizational analogy, the MIB defines the information conveyed between the branch office and the main office.
- A *data definition language*, known as SMI (Structure of Management Information). SMI defines the data types, an object model, and rules for writing and revising management information. MIB objects are specified in this data definition language. In our human organizational analogy, the SMI is used to define the details of the *format* of the information to be exchanged.
- A *protocol*, *SNMP*. SNMP is used for conveying information and commands between a managing entity and an agent executing on behalf of that entity within a managed network device.
- *Security and administration capabilities*. The addition of these capabilities represents the major enhancement in SNMPv3 over SNMPv2.

The Internet network management architecture is thus modular by design, with a protocol-independent data definition language and MIB, and an MIB-independent protocol. Interestingly, this modular architecture was first put in place to ease the transition from an SNMP-based network management to a network management framework being developed by ISO, the competing network management architecture when SNMP was first conceived—a transition that never occurred. Over time, however, SNMP’s design modularity has allowed it to evolve through three major revisions, with each of the four major parts of SNMP discussed above evolving independently. Clearly, the right decision about modularity was made, even if for the wrong reason!

In the following subsections, we cover the four major components of the Internet-Standard Management Framework in more detail.

9.3.1 Structure of Management Information: SMI

The **Structure of Management Information, SMI** (a rather oddly named component of the network management framework whose name gives no hint of its functionality), is the language used to define the management information residing in a managed-network entity. Such a definition language is needed to ensure that the syntax and semantics of the network management data are well defined and unambiguous. Note that the SMI does not define a specific instance of the data in a managed-network entity, but rather the language in which such information is specified. The documents describing the SMI for SNMPv3 (which rather confusingly, is called SMIv2) are [RFC 2578; RFC 2579; RFC 2580]. Let’s examine the SMI in a bottom-up manner, starting with the base data types in the SMI. We’ll then look at how managed objects are described in SMI, then how related managed objects are grouped into modules.

SMI Base Data Types

RFC 2578 specifies the basic data types in the SMI MIB module-definition language. Although the SMI is based on the ASN.1 (Abstract Syntax Notation One) [ISO X.680 2002] object-definition language (see Section 9.4), enough SMI-specific data types have been added that SMI should be considered a data definition language in its own right. The 11 basic data types defined in RFC 2578 are shown in Table 9.1. In addition to these scalar objects, it is also possible to impose a tabular structure on an ordered collection of MIB objects using the SEQUENCE OF construct; see RFC 2578 for details. Most of the data types in Table 9.1 will be familiar (or self-explanatory) to most readers. The one data type we will discuss in more detail shortly is the OBJECT IDENTIFIER data type, which is used to name an object.

SMI Higher-Level Constructs

In addition to the basic data types, the SMI data definition language also provides higher-level language constructs.